

Frontend Integration Guide: Step 1 (Norms) & Step 2 (Vacancy)

Status: Final Production Version (AI-Centric & SRS Aligned)

Scope: Faculty Requirement Generation (Step 1) and Vacancy Identification (Step 2).

1. AI Business Objectives (SRS 7.6.3)

1.1 Faculty Requirement Calculator (AI-Assisted)

The system uses AI-assisted rule validation to:

- Calculate faculty requirements based on intake, course norms, and faculty-student ratios.
- Compare current requirements with historical utilization data.
- Flag abnormal variations for review.
- **Outcome:** AI generates a **"Suggested Requirement Summary"**, while final approval remains with the Directorate.

1.2 Faculty Allocation & Vacancy Identification

AI assists by:

- Analyzing existing faculty strength and workload.
 - Identifying optimal CHB vacancy distribution.
 - Highlighting underutilized or overloaded courses.
 - **Outcome:** The system presents **AI-recommended vacancy lists with justification**, enabling faster and more accurate approvals.
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2. Step 1: Norms & Requirement AI

2.1 Norm Configuration

Admin defines the rules that govern the entire institution. These 5 fields are critical for the AI calculation:

- `faculty_student_ratio`: Integer (e.g. **15** for 1:15).
- `max_age`: Integer (e.g. **38**).
- `workload_hours_per_week`: Integer (e.g. **18**).
- `min_qualification`: String (e.g. **"M.E./M.Tech"**).
- `grade_requirement`: String (e.g. **"First Class"**).

2.2 Requirement Validation (AI-Assisted)

Endpoint: `POST /api/requirements/validate` **UI Action:** Call this on the "Review Requirement" screen to generate the **"Suggested Requirement Summary"**.

- **Actual LLM Output:**

```
{
  "ai_summary": "Based on current intake, requirement appears NORMAL...",
  "confidence_score": 0.95,
  "insights": ["Faculty requirement changed by 5% vs last year"],
  "anomalies": [...]
}
```

3. Step 2: Vacancy Intelligence (Multi-Norm Engine)

The system identifies vacancies not just by numbers, but by **Norm Compliance**.

3.1 Unified AI Analysis Endpoint

Endpoint: `POST /api/vacancies/ai-analysis` **UI Action:** Use this to show the "AI-recommended vacancy list with justification".

- **Actual LLM Output:**

```
{
  "system_vacancy": 5,
  "ai_analysis": {
    "ai_suggested_vacancy": 6,
    "norm_compliance_score": 0.75,
    "justification": "System found 2 faculty members over 38 years old and 1 with only a B.E. (M.Tech required).",
    "overloaded": ["Thermal Engineering Lab"],
    "underutilized": ["Part-time staff"],
    "insights": ["High growth in student intake requires 1 additional CHB"]
  }
}
```

3.2 UI Display Rules:

1. **Requirement Summary:** Display the `ai_summary` in a prominent header on the Requirement Approval page.
2. **Vacancy Recommendation:** Show `ai_suggested_vacancy` next to the system count.
3. **Justification Card:** Always show the `justification` string to provide transparency for "faster and more accurate approvals."
4. **Workload Insights:** List the `overloaded` and `underutilized` areas to help the Principal allocate duties.

4. Anomaly Feedback Loop

Highlight rows in the faculty list where anomalies are present:

Flag	Logic Trigger	UI Tooltip
UNDER_QUALIFIED	Faculty degree < min_qualification	"Does not meet M.Tech requirement."
OVER_AGE	Current Age > max_age	"Exceeds the 38-year DTE limit."

5. Developer Notes

- **Loading Indicators:** Show an "AI Analyzing..." spinner during the 2-4 second latency.
- **Fallbacks:** If AI fails, show the `system_calculation` and a fallback note.
- **PDF Export:** Use the "Markdown PDF" extension in VS Code to share this guide as a PDF.